

I. Executive summary

Injury¹ has become a major cause of death and disability worldwide. Organized approaches to its prevention and treatment are needed. In terms of treatment, there are many low-cost improvements that could be made to enhance the care of injured persons. The goal of the *Guidelines for essential trauma care* is to promote such low-cost improvements. These guidelines seek to set achievable standards for trauma treatment services which could realistically be made available to almost every injured person in the world. They then seek to define the resources that would be necessary to assure such care. These include human resources (staffing and training) and physical resources (infrastructure, equipment and supplies). By more clearly defining such services and resources, we hope these guidelines will facilitate the strengthening of trauma treatment services worldwide. The basic premise of these guidelines is that improvements in organization and planning can result in improvements in trauma treatment services and hence in the outcome of injured persons, with minimal increases in expenditures.

The authors of the guidelines have developed a series of resource tables for essential trauma care that detail the human and physical resources that should be in place to assure optimal care of the injured patient at the range of health facilities throughout the world, from rural health posts whose staff do not have training as doctors, to small hospitals staffed by general practitioners (known as GP-staffed hospitals), to hospitals staffed by specialists (specialist-staffed hospitals), to tertiary care centres. They also take into account the varying resource availability across the spectrum of low- and middle-income countries.² Finally, a series of recommendations is made on methods to promote such standards including training, performance improvement, trauma team organization and hospital inspection. The resource tables and associated recommendations are

¹ The terms “injury” and “trauma” are used interchangeably throughout these guidelines.

² In these guidelines, mention is made of the categorization of countries by economic level according to the criteria of the World Bank (www.worldbank.org/data/countryclass/countryclass.html), based on 2002 gross national income (GNI) per capita: low-income, US\$735 or less; lower-middle-income, US\$736–2935; upper-middle-income US\$2936–9075; and high-income, US\$9076 or more.

intended to provide a template to assist individual countries in organizing and strengthening their own trauma treatment systems. It is anticipated that the template will be adapted to suit local circumstances.

These recommendations have been drawn up in collaboration between the following parties:

- the Department of Injuries and Violence Prevention (VIP) of the WHO
- members of the Working Group for Essential Trauma Care of the International Association for the Surgery of Trauma and Surgical Intensive Care (IATSIC), which is an integrated society within the broader International Society of Surgery/Société Internationale de Chirurgie (ISS/SIC);
- representatives of other organizations and other departments of WHO, such as the Department of Essential Health Technologies, which are involved in developing training materials on essential surgical care and trauma; and
- trauma care clinicians from Africa, Asia and Latin America.

The guidelines are written in a style oriented primarily towards health care planners and administrators. In many circumstances, these may include clinicians who are involved in the care of injured patients but also have administrative duties. The guidelines are also intended for use by clinicians who might use them to demonstrate to planners and administrators the need to make improvements to the resources that are available for trauma care. Thus, the target audience for these guidelines includes planners in ministries of health, hospital administrators, nursing service directors, medical service directors and clinicians, both individually and collectively, through organizations such as societies of surgery, anaesthesia, traumatology and other disciplines that deal with the injured patient. Broadly construed, these guidelines are of relevance to anyone involved in planning trauma care services or anyone who might wish to promote improvements in the care of the injured in their country.

2. Orientation to the Essential Trauma Care Project

2.1 Goals of the Essential Trauma Care Project

In working towards decreasing the burden of death and disability from injury, a spectrum of activities needs to be considered, ranging from surveillance and basic research to prevention programmes, to trauma management. Large gains are to be made in prevention, and hence a major emphasis should be placed on this approach. There are also major gains to be made by addressing treatment. That is to say, low-cost initiatives can help to reinforce current trauma treatment systems worldwide and by so doing help to reduce the overall burden from injury. The goals of the Essential Trauma Care (EsTC) Project are to identify and promote such inexpensive ways of reinforcing trauma treatment worldwide.

The EsTC Project seeks to accomplish this overall goal by better defining what essential trauma treatment services should realistically be made available to almost every injured person worldwide. The project then seeks to develop ways of assuring the availability of these services by reinforcing inputs of: (1) human resources (training and staffing); and (2) physical resources (supplies and equipment). These inputs are outlined in the form of a template, referred to as the EsTC resource matrix. It is intended that this template and the entire manual should be used as a guide for those planning trauma treatment services for their countries or areas.

The following chapter is intended as an introduction to the concept and development of the EsTC Project. It outlines the current disparities in outcome between persons injured in high-income countries and those injured in low- and middle-income countries. It briefly reviews some of the difficulties facing trauma care in developing countries which may account for existing disparities and which are the focus of efforts to improve organization and reinforce services. It reviews the foundations we have on which to build in our efforts to improve trauma treatment services, including essential health service programmes for other diseases and efforts to improve trauma care in individual countries. It introduces the premise that improvements in the organization of trauma treatment services may represent a cost-effective way of improving the delivery and outcome of such services, and presents evidence to this effect. It then summarizes the process by which this manual was developed and written, and suggests how it should

ultimately be used. Finally, it addresses the overlap between the EsTC Project and other international health activities, including WHO activities.

2.2 Disparities in outcome of trauma

There are notable disparities in mortality rates for injured patients around the world. For example, one study looked at the mortality rates for all seriously injured adults (injury severity score of 9 or more) in three cities, in countries at different economic levels. The mortality rate (including both pre-hospital and in-hospital deaths) rose from 35% in a high-income setting to 55% in a middle-income setting, to 63% in a low-income setting (1). Considering only patients who survive to reach the hospital, a similar study demonstrated a six-fold increase in mortality for patients with injuries of moderate severity (injury severity score of 15–24). Such mortality increased from 6% in a hospital in a high-income country to 36% in a rural area of a low-income country (2).

In addition to an excess mortality, there is a tremendous burden of disability from extremity injuries in many developing countries (2, 3). By comparison, head and spinal cord injuries contribute a greater percentage of disability in high-income countries (4). Much of the disability from extremity injuries in developing countries should be eminently preventable through inexpensive improvements in orthopaedic care and rehabilitation.

In part, the improved survival and functional outcome among injured patients in developed countries comes from high-cost equipment and technology. Unfortunately, much of this may be unaffordable to the average injured person in the world for the foreseeable future. However, much of the improvement in patient outcome in higher-income countries has come from improvements in the organization of trauma care services (5–9). Improvement in the organization of trauma services should be achievable in almost every setting and may represent a cost-effective way of improving patient outcomes. This is the basis of the Essential Trauma Care Project. Before going into further detail about the plans for the project itself, it is useful to briefly examine some of the difficulties facing trauma care in developing countries that might be improved upon in an inexpensive fashion through improvements in organization and planning.

2.3 Elements of trauma care to reinforce

This section provides brief examples of the deficiencies in resources and organization that could be targeted by an Essential Trauma Care Project. These are considered in three broad categories.

Human resources: staffing and training

In most developing countries, little consideration has been given to optimizing the training of medical and nursing staff for the care of injured patients. This applies

to both urban and rural environments. For example, hospitals in rural areas along major roads in Ghana receive large numbers of casualties, yet are often staffed by general practitioners (GPs) and nurses with no specific training in trauma care. Moreover, 30% of the GPs in these hospitals have not had a surgery rotation during their training as house officers (internship) (10). Inexpensive but effective solutions to such problems might include: (i) country-wide plans to encourage those staff with more experience in trauma care to go to such hospitals; and (ii) provision of continuing education courses on trauma care for GPs and nurses in high-volume trauma hospitals. Pilot programmes for such training have demonstrated improvements in trauma care in a variety of environments, including Ghana and Trinidad (10–14).

Physical resources: infrastructure, equipment and supplies

Many hospitals lack important equipment, some of which is inexpensive. For example, in Ghana, it was found that of 11 rural hospitals along major roads, none had chest tubes and only 4 had emergency airway equipment (11). These items are vitally important for the treatment of life-threatening chest injuries and airway obstruction, major preventable causes of death in trauma patients. Such equipment is inexpensive and much of it is reusable. The main reason for the absence of such vital equipment is a lack of organization and planning, rather than resource restrictions (10, 11). In Mexico, a lack of adequate capabilities for the repair of the one CT (computerized tomography) scanner at a major urban trauma centre resulted in prolonged periods of breakdown, which greatly impaired the care of head-injured patients (15). Programmes to assure the supply and maintenance of trauma-related equipment, appropriate to the specific circumstances of the given country, could help to address deficiencies of inexpensive but high-yield resources.

Process: organization and administration

In addition to assuring adequate supplies, improved administration could also assist in appropriate utilization. In a study in Ghana, it was found that, even at an urban teaching hospital, there were significant problems with the process of trauma care. There were prolonged times to emergency surgery, with a mean time of 12 hours between arrival at the casualty ward and start of the procedure. There was also low utilization of chest tubes, even though they were physically available in the hospital. Only 0.6% of all trauma admissions had a chest tube inserted, in comparison with 6.3% at a hospital in a developed country. In addition to addressing such items individually, this study identified the implementation of basic quality improvement programmes (medical audit) as a possible way of addressing many such problems in the process of trauma care simultaneously (16).

The Essential Trauma Care Project is viewed as a way of comprehensively addressing such difficulties and deficiencies. Any such effort must build on existing foundations, which are considered next.

2.4 Foundations on which to build

In initiating the Essential Trauma Care Project, there are two sets of activities upon which the authors seek to build: essential health services and efforts to improve trauma care in individual countries.

Essential health services

The World Health Organization (WHO) and others involved in international health have made considerable progress in improving health in the array of developing countries by advancing the concept of essential services. These are services which are low in cost and high in yield, which target major health problems, and which can realistically be made available to almost everyone in a given population. Programmes for these services have included defining, refining and promoting these services, as well as providing technical input to countries to help improve their capacity to deliver the services. Examples of essential service programmes include: the Essential Drugs Programme, the Expanded Programme on Immunization, the Global Tuberculosis Programme and the Safe Motherhood Initiative.

Efforts to improve trauma care in individual countries

As an example of this, the American College of Surgeons (ACS) Committee on Trauma has significantly advanced the care of the injured in the United States and Canada by creating and promulgating the Advanced Trauma Life Support course (ATLS) and by the publication of *Resources for optimal care of the injured patient* (17). This 100-page book contains guidelines for what hospitals at varying levels should have in place in terms of resources such as staffing, continuing education, supplies, equipment, administrative functions and quality assurance programmes. Basically, it has applied an essential services approach. Equally important to the existence of these standards is the fact that they have been enforced through a programme of trauma centre verification, as described in section 2.5. Similar organized measures have been successfully implemented in many other high-income countries (18–21).

In addition to efforts to improve care at individual hospitals, progress has been made by addressing the entire spectrum of the development of systems for trauma management. This involves political jurisdictions designating hospitals to fill the roles of trauma centres at varying levels of complexity, ranging from large urban centres to small rural hospitals. It also implies planning of emergency medical services, pre-hospital triage, transfer criteria and transfer arrangements between hospitals.

It should also be noted that every country has capable, dedicated individuals working to reduce the burden of injury, both by planning and administration and by the direct provision of clinical services. The Essential Trauma Care Project and this manual are meant to assist and reinforce such work.

The Essential Trauma Care Project seeks to blend the perspectives of the two foundations noted above. The project is based on the premise that improved organization of trauma care services can be a cost-effective way of improving care of the injured worldwide. Before going further, evidence for the effectiveness of such improved organization is presented.

2.5 Evidence of the impact of better organization of trauma services

Almost all of the evidence of the effectiveness of improvements in the organization of trauma care services comes from developed countries. In most cases, the better organization comes in the form of two related activities: (1) verification of trauma services through hospital inspections; and (2) planning of systems for trauma management. Verification applies to a review of individual facilities as regards their provision of a variety of items, including human resources (e.g. availability of personnel with certain qualifications), physical resources (equipment and supplies) and administrative and organizational functions, such as quality improvement. The planning of systems for trauma management implies several integrated functions, including political jurisdictions designating which hospitals are to fill the roles of trauma centres at varying levels of complexity, ranging from large urban trauma centres to small rural hospitals and clinics. It also implies the planning of mobile emergency medical services, pre-hospital triage (to determine which patients should go to which types of designated facilities), transfer criteria and transfer arrangements between hospitals.

There is considerable evidence that political jurisdictions that improve the organization of trauma services benefit from reduced trauma mortality, in comparison with similarly resourced jurisdictions that do not. Such evidence comes from panel reviews of preventable deaths, hospital trauma registry studies and population-based studies (7, 22, 23). Most studies confirm a reduction in mortality with the improved organization provided by a system for trauma management. For example, panel reviews show an average reduction in medically preventable deaths of 50% after the implementation of a system for trauma management. Likewise, population-based studies and trauma registry studies show a fairly consistent 15–20% or greater reduction in mortality for better organized systems, compared with either the same systems prior to improvements in organization or to other less organized systems (6, 7, 23).

In one of the best series of studies done on this topic, Nathens et al. used population-based data to examine the effects of planning of systems for trauma

management in all of the 50 states of the United States. They looked at mortality rates, adjusting for several potential confounding variables, including traffic laws and other safety measures. They demonstrated an 8% reduction in mortality for those states with systems for trauma management. It should be mentioned that this figure represents overall trauma mortality, including persons dead at the scene, before any chance of medical treatment. In comparison, the larger reductions in mortality mentioned above represented changes in mortality for patients surviving to reach the hospital. Especially notable in Nathens' study was the finding that the effect of a system for trauma management was not usually evident until 10 years after its initial enactment and reached a maximum at 16 years (8, 9).

When considering the relevance of these findings to the potential utility of similar organizational efforts in developing countries, it is important to note that the above improvements were mostly witnessed in comparison with environments with the same levels of resources. The enactment of an organized system for trauma management usually required inputs of resources that were fairly small in comparison with the overall cost of the existing system of care itself. The system for trauma management itself did require the funds for increased organization and occasional extra inputs of resources to bring institutions up to standards for verification (6). The Essential Trauma Care Project is based on making similar improvements in organization and planning that are inexpensive in comparison with the cost of the existing treatment system itself.

2.6 Development process for essential trauma care

In developing the *Guidelines for essential trauma care*, the authors sought to define inexpensive, feasible, minimal standards that would be applicable virtually everywhere in the world. They also sought to identify ways of reinforcing existing systems of trauma care in all locations in the world, including the spectrum of conditions found in both low- and middle-income countries.

In this process, the authors developed a list of medical goals that should be feasible for most injured persons everywhere. These can be viewed as the "needs of the injured patient" (chapter 3). In order to assure the achievement of such goals, the inputs of human and physical resources must be utilized in an optimal process. To this end, the authors have developed a template for the resources that are needed. These are described in chapters 4 and 5. These are based conceptually on a similar template developed for the book, *Resources for optimal care of the injured patient* (17). However, they are significantly amended to reflect the realities of low- and middle-income countries.

The authors envision that the template presented in this manual will be used as a guide for those planning trauma treatment services across a wide spectrum of low- and middle-income countries. These guidelines will hopefully be of relevance to planners in ministries of health, to hospital administrators, to nursing

service directors and to clinicians, both individually and collectively, through organizations such as societies of surgery, anaesthesia, traumatology and other disciplines that deal with the injured patient. These groups constitute the target audience for the guidelines.

The use of the template will require adaptation for use in individual countries. The authors envision that this would ideally occur in a collaborative, consensus-driven process, with input from clinicians involved in the care of injured patients, officials in ministries of health, and health service administrators. The category of clinicians would include both those who are primarily devoted to the field of traumatology and generalists such as general surgeons, anaesthetists and emergency physicians. In countries in which a large percentage of rural trauma is handled by general practitioners and non-doctor providers of health care, these groups should be included as well, to assure the feasibility of recommendations for rural areas.

As part of the process of national level adaptation and implementation, there could reasonably be a role for a Needs Assessment of trauma treatment services nationwide. This would involve a survey of the capabilities of the range of facilities caring for injured persons. It would help to identify the potential deficiencies in human resources, physical resources and organization that would be targeted by subsequent improvements.

2.7 Overlap with other activities

The guidelines provided by this manual deal primarily with facility-based trauma care. A very important component of trauma care with which it does not deal is pre-hospital care, or mobile emergency medical services (EMS). The authors acknowledge the great importance of this topic, especially as most trauma deaths in almost all countries occur outside of hospital (1). A WHO publication addressing this area of care is in preparation. Furthermore, there is no hard and fast line between pre-hospital care and hospital-based care, nor between stabilization and definitive care. For example, rural clinics whose staff are not doctors are often the first source of care for injured patients in their communities. These are intermediate between pre-hospital and hospital-based in character, as regards trauma care. Hence, for purposes of these guidelines, the authors have decided not to directly address mobile pre-hospital care, which is usually categorized as EMS. They have decided to consider care at all fixed facilities, whether these be clinics or actual hospitals.

Likewise, the guidelines in this manual deal exclusively with the care of injured patients. It is acknowledged that, except in rare circumstances, the human and physical resources in place will also be utilized for other health problems, including medical, obstetric and other surgical problems. Hence, there is a need to integrate the guidelines with other emergency services. It is hoped that the upgrading

of trauma care resources will have the secondary effect of improving other aspects of emergency medical care as well. To increase the probability of this outcome and to decrease the probability that such recommendations might rather have a detrimental effect on other aspects of health care, the authors have sought the input of other departments of the WHO and other groups that are involved in emergency care and health care in general. In other words, these guidelines are vertically oriented with respect to trauma care. However, the authors have taken measures to assure that they are horizontally integrated into the broader functioning of the health care system. In this regard, the guidelines have been developed with the input of persons involved in the following WHO programmes and departments: Essential Drugs and Medicines Policy, Management of Non-communicable Diseases, Disability and Rehabilitation, Essential Health Technologies, Diagnostic Imaging and Laboratory Technology, and Child and Adolescent Health.

The *Guidelines for essential trauma care* are not clinical algorithms. They represent an attempt to provide sufficient resources that such algorithms can be carried out effectively and safely. For further details of pertinent emergency algorithms, the reader is referred to the *Integrated management of adolescent and adult illness*, which is in preparation by WHO (http://www.who.int/gtb/publications/whodoc/imai/cds_stb_2003_22.pdf).

The guidelines are concerned only with the care of the injured, and hence with secondary and tertiary prevention of injury-related death and disability. Obviously, primary injury prevention is of great importance. Clinicians of all types should be strongly encouraged to become involved in promoting primary injury prevention. For further details, readers are referred to the related activities and publications of the WHO's Injuries and Violence Prevention Department (www.who.int/violence_injury_prevention).

Finally, the ultimate utility of these guidelines consists in their ability to actually enact improvements in the process of trauma care that lead to decreases in mortality and disability due to trauma. In order to monitor the success or failure of such efforts and to be able to make modifications when success has not been achieved, it is necessary to have reliable and timely sources of information on the incidence of injury and its outcome. This implies some form of surveillance. For further information on this, the reader is referred to the WHO publication: *Injury surveillance guidelines* (24).

3. Essential trauma services: needs of the injured patient

This section contains a list of those services which the authors feel are essential to prevent death and disability in injured patients. They might be considered as the “needs of the injured patient.” These can be categorized into three broad sets of needs:

1. Life-threatening injuries are appropriately treated, promptly and in accordance with appropriate priorities, so as to maximize the likelihood of survival.
2. Potentially disabling injuries are treated appropriately, so as to minimize functional impairment and to maximize the return to independence and to participation in community life.
3. Pain and psychological suffering are minimized.

Within these three broad categories, there are several specific medical goals that are eminently achievable within the resources available in most countries.

- Obstructed airways are opened and maintained before hypoxia leads to death or permanent disability.
- Impaired breathing is supported until the injured person is able to breathe adequately without assistance.
- Pneumothorax and haemothorax are promptly recognized and relieved.
- Bleeding (external or internal) is promptly stopped.
- Shock is recognized and treated with intravenous (IV) fluid replacement before irreversible consequences occur.
- The consequences of traumatic brain injury are lessened by timely decompression of space occupying lesions and by prevention of secondary brain injury.
- Intestinal and other abdominal injuries are promptly recognized and repaired.
- Potentially disabling extremity injuries are corrected.
- Potentially unstable spinal cord injuries are recognized and managed appropriately, including early immobilization.
- The consequences to the individual of injuries that result in physical impairment are minimized by appropriate rehabilitative services.

- Medications for the above services and for the minimization of pain are readily available when needed.

The precise procedures that can optimally be applied to achieve these goals, as well as the human and physical resources needed to optimally carry out these procedures, will vary across the spectrum of economic resources of the nations of the world and the geographic location of the facilities concerned. However, these goals should be achievable for most injured patients in most locations.

The provision of these services should not be dependent on ability to pay. Hence, cost recovery schemes, necessary though they may ultimately be, should not preclude the provision of initial emergency care nor of critical elements of definitive care.

4. Inputs needed to achieve essential trauma services

4.1 Overview

The goals outlined in the previous chapter depend on the provision of specific items of physical examination, diagnostic tests, medications and therapeutic procedures. Likewise, the ability of the health system to provide these items depends on the inputs of human resources (training and staffing) and physical resources (infrastructure, equipment and supplies). The following sections of this manual outline those resources which the authors feel are essential to the provision of essential trauma services. These resources are outlined in the form of the resource matrix for essential trauma care (Tables 1–14).

4.2 Resource matrix: introduction

The resource matrix for essential trauma care contains brief descriptions of the resources that need to be available for the provision of specific categories of care at different levels of the health care system. A specific matrix is derived for each of 14 categories of care, such as airway, shock, head injuries, extremity injuries and rehabilitation. These include both initial emergency management and long-term definitive care.

4.3 Resource matrix: necessary elements of trauma care

On the vertical axis of each matrix are listed the specific elements of trauma care that are needed. These are divided into two categories: (1) knowledge & skills and (2) equipment & supplies. See Table 1 as an example.

Knowledge and skills imply that the staff (medical, nursing and others) have the requisite training to perform such diagnostic and therapeutic activities safely and successfully. This implies not only the requisite training in their basic education (school and postgraduate training), but also continuing education to maintain these skills. Training issues are comprehensively addressed in chapter 6 of this manual.

Equipment and supplies imply that these items are available to all who need them, without consideration of ability to pay, especially in true life-threatening emergencies. This implies not only having them physically present in the facility but having them readily available on an ongoing basis; where appropriate, 24 hours

5. Guidelines for essential trauma care

For each of the following 14 categories of trauma care, a summary resource matrix is provided, preceded by a brief explanation of the rationale used in determining which elements of care are to be considered essential or desirable. An explanation of the organization of the resource matrices is provided in section 4. The final category (section 5.15) deals with special considerations for the care of injured children.

5.1 Airway management

Airway management is one of the key components of emergency care. Its primary objective is to diagnose an obstructed or potentially obstructed airway, to clear the obstruction and keep the airway patent. No medical emergency, short of a complete cardiopulmonary arrest, is more immediately life-threatening than the loss of an adequate airway. Failure to adequately manage airway patency and ventilation has been identified as a major cause of preventable death in trauma (27–30).

In the initial assessment and management of any critically ill patient, the airway, breathing and circulation (ABC) are the first steps. The response to any acutely ill or injured patient must be met using a systematic approach, with the airway being the first priority. If any abnormalities are detected, measures to intervene are instituted immediately. The skills to assess a patient for obstruction of the airway, to establish and maintain a patent airway, and to ensure adequate ventilation and oxygenation of the patient, are therefore essential.

At all levels of the health care system, it is deemed essential that health care personnel know the signs of airway obstruction and are skilled in manual manoeuvres to keep an airway patent while maintaining cervical spine protection. The probability of success of airway management is increased by the provision of specific equipment and the skills to utilize it properly and safely. This includes equipment and skills for basic airway management, including oral or nasal airway, suction and bag–valve–mask. These are deemed essential at hospital-level facilities. The probability of success in airway management is further increased by the provision of equipment and skills for advanced airway

management, including endotracheal intubation and cricothyroidotomy. These are deemed essential at specialist-staffed hospitals. At each level, the probability of success in airway management is increased. However, both the need for resources (both equipment and training) and the potential for harm are increased. Environments in which more resources are available, and/or in which there is a greater than average trauma volume in specific facilities, may wish to change the “desirable” designation to essential at some or all facilities of a given level.

At whatever level of the health care system it is decided to provide advanced airway capabilities (e.g. endotracheal intubation with or without cricothyroidotomy, with or without tracheostomy—see end of section 5.1),¹ several safety assurances should be in place. This includes the provision that the equipment is readily available in whatever area receives the injured patient (casualty ward or emergency department). This is aided by having the necessary equipment in pre-assembled packs (where appropriate and feasible), stocked in the emergency area. Safety assurance also mandates that staff performing the various procedures be adequately trained to perform them successfully, with an acceptable rate of complications. This includes both the training received in basic education (e.g. medical or nursing or other professional school) and whatever continuing education might be required to maintain the skills. Given the potential for harm (in the form of oesophageal intubation) with advanced airway management, inexpensive materials to assist in the clinical assessment of endotracheal tube placement should be provided whenever endotracheal intubation is performed. This includes principally an oesophageal detector device (either bulb or syringe). Finally, given this potential for harm associated with advanced airway manoeuvres, a quality assurance mechanism should be in place to track adverse events such as oesophageal intubations. Further details can be found in the chapter on quality assurance.

By way of further explanation of some of the equipment listed in the table, suction is an extremely important component of airway management. It can be provided at a low cost by manual and foot pump devices. These should be considered essential in any hospital. Likewise, a stiff suction tip (Yankauer or equivalent) is an essential component of an adequate suction set up. The term “basic trauma pack” implies a kit with a few basic instruments and supplies, including a scalpel, clamps, scissors, gauze, suture, syringe and needles. These represent a component of the minimum of physical resources needed to perform certain smaller procedures in the casualty ward/emergency department setting. Such pro-

¹ Cricothyroidotomy is generally considered to be the surgical airway of choice in emergency situations and can be performed in several seconds. If needed for a prolonged period, it is usually converted to a tracheostomy after a few days.

5.2 Breathing—Management of respiratory distress

The ability to assess a patient for respiratory distress and adequacy of ventilation is essential at all levels of the health care system. This applies both to those initially evaluating the patient and to those who are providing definitive care. The only resources required for this function are training and a stethoscope. If no other resources are available at the level in question, it is anticipated that respiratory distress would usually constitute grounds for referral to the next highest level of the system.

Capabilities for the administration of oxygen to trauma patients in respiratory distress are essential at all hospital facilities. This would be useful at all levels of the health system. It is recognized that this is currently beyond the realm of feasibility for most primary health care clinics with non-medical staff in low-income countries. However, facilities that receive a moderate volume of seriously injured patients (e.g. those located along busier roadways), especially in middle-income settings, might realistically be supplied with oxygen. The capability for administration of oxygen implies both health care providers capable of understanding the indications for its use and equipment and supplies to administer it in a timely fashion to trauma patients in respiratory distress. In most cases, this implies that the physical resources are present in the area where acute trauma patients are initially received. WHO's Department of Essential Health Technologies (EHT) is developing guidelines for the use of oxygen and related training and equipment (<http://www.who.int/ehs>).

The recognition of tension pneumothorax, its primary treatment by needle thoracostomy and definitive treatment by tube thoracostomy are essential at all hospital-level facilities that handle trauma. This implies sufficient training of the principal caregiver in the diagnosis of tension pneumothorax and in the safe performance of the relevant procedures. It also implies the ready availability of the needed materials, including a basic trauma pack, chest tubes and underwater seal drainage bottles.

At the basic health care level, capabilities for the recognition of tension pneumothorax and temporary relief with a needle thoracostomy could be considered desirable in settings with the possibility of rapid evacuation to a site of definitive treatment. This would usually imply the existence of an EMS system.

Recognition of the presence of a sucking chest wound and the ability to apply a three-way dressing for immediate treatment is deemed essential at all levels.

Assessment of the adequacy of supplemental oxygen is based primarily on clinical examination. Supplemental laboratory measurements (arterial blood gas concentration) and monitoring (oxygen saturation through pulse oximetry) provide further useful information. However, their cost prevents them from being deemed

5.3 Circulation—Management of shock

Assessment of shock and control of external haemorrhage

The ability to assess a patient for the presence of shock is essential at all levels of the health care system. The only resources required for this function are a clock or watch with second hand, a stethoscope and blood pressure (BP) cuff, and the relevant training. Training should include visual/manual assessment of circulation, including pulse, venous filling and skin temperature. If no other resources are available at the level in question, it is anticipated that shock would usually constitute grounds for referral to the next highest level of the system.

Control of external haemorrhage through manual pressure and through the application of a pressure dressing is essential at all levels of the health care system. The only resources required are training and sufficient gauze bandages. These should be sterile whenever possible. Given the limited facilities for sterilization at most basic facilities, clean bandages should be considered essential at this level. Sterile bandages are considered essential at all hospital levels. Training regarding indications for the use of arterial tourniquets in extreme situations, as well as understanding the potential complications and the need for removal under controlled circumstances within several hours, is essential at all levels. Splinting of fractured extremities as a means of decreasing internal haemorrhage is essential at all levels.

More advanced, externally applied haemorrhage control measures include wrapping for potential pelvic fractures and deep interfascial packing for complicated wounds, such as landmine and machete wounds. These are deemed desirable at the basic level and essential at all hospital levels.

Fluid resuscitation

Capabilities for fluid resuscitation include the equipment, the fluids themselves and the skills to administer them, monitor the response (including accurately monitoring fluid intake and output) and treat potential complications. The insertion of peripheral intravenous (IV) lines, percutaneously and by cutdown, and the use of crystalloid are deemed essential at all hospital levels. These are desirable at even the most basic levels at which seriously injured patients are seen. This is especially the case in locations where the basic level is at such a distance that evacuation to a higher level of care will entail a delay of several hours or more. The insertion of central lines (including the lines themselves and the expertise to insert them safely) is deemed essential at the upper two hospital levels and desirable at GP-level hospitals. However, in the setting of GP hospitals, given the potential for complications, the insertion of central lines should really only be considered for emergency situations in which access cannot be achieved by any other means.

The use of colloid is deemed desirable at higher hospital levels and should comply with existing WHO Essential Drug Programme guidelines. Intraosseous lines for children, especially for those under 5 years, are deemed essential at all hospital levels. Formal intraosseous needles would be ideal, but the ability to establish intraosseous access using any suitable large-bore metal needle (e.g. spinal needle) is acceptable.

Capabilities for blood transfusion are deemed essential at all hospital levels. A formal blood bank is best. However, if not available, capabilities for immediate donation and administration of fresh whole blood are acceptable. Such capabilities are also needed for the treatment of obstetric haemorrhage and severe anaemia. Most GP-level hospitals need to provide transfusions for these indications as well as for trauma. A small minority of GP-level hospitals might not be expected to have such capabilities. These would include smaller facilities in less remote areas, with easy access to referral centres. Such facilities might be considered to more closely represent the basic level than the GP-level hospitals considered in these guidelines.

Any time that blood is administered, there should be capabilities to assure its safety, including screening for HIV, hepatitis B and C, and other blood borne diseases, depending on the geographic area. Use of blood should follow existing WHO Blood Transfusion Safety guidelines and associated national policies (http://www.who.int/ehd/Main_areas_of_work/BTS/BTS.htm). The use of blood also implies that it is being ordered by a clinician who knows the indications for transfusion in a trauma patient and is capable of recognizing and treating the potential complications of transfusion, monitoring the patient's response to transfusion and other fluid resuscitation, and assessing the patient for continued bleeding and the need for surgical intervention.

Monitoring

The capability for monitoring a patient in shock for response to fluid therapy is deemed essential at all hospital levels. This includes an understanding of the stages of haemorrhagic shock. It also includes the following basic equipment: clock or watch with second hand, stethoscope, blood pressure cuff and urinary catheter with collection bag (or improvised equivalent). The following monitoring devices add utility, but also cost, and hence are deemed desirable, depending on the hospital level (see table): electronic cardiac monitoring, monitoring of central venous pressure and right-heart catheterization with monitoring of pulmonary capillary wedge pressure.

Laboratory tests assist in the assessment of the presence of shock, the degree of bleeding and response to resuscitation. Haematocrit or haemoglobin monitoring is deemed essential at all hospital levels. Measurements of electrolytes (sodium, potassium, chloride, bicarbonate), blood urea nitrogen, creatinine,

and because of the cost needed to adequately train both medical and nursing staff in the safe administration of pressors by continuous IV drip. The ability to treat septic shock with antibiotics, fluid resuscitation and other supportive care is deemed essential at all hospital levels. Appropriate treatment of source of infection and underlying injury is covered further in several of the following sections.

Many patients in shock develop an ileus and are at risk of vomiting and aspiration of gastric contents. Hence, the availability of nasogastric (NG) tubes and the skills to insert them are deemed essential at all hospital levels.

Hypothermia is a frequent complication of shock. The ability to recognize this and to treat it with external rewarming is deemed essential at all levels. Capabilities to provide warmed fluids and gases, as well as other means of rewarming the body core, such as lavage via urinary catheters, nasogastric tube or intraperitoneal catheter, are considered desirable at all hospital levels.

Weighing scales for children, in order to more accurately calculate fluid requirements, are considered essential at all hospital levels and desirable at the basic level.

5.4 Management of head injury

Sections 5.1–5.3 described in detail the specific skills, equipment and supplies needed to treat the immediately life-threatening injuries addressed in the initial evaluation and management. For more complicated issues, these guidelines provide less specific detail. Details of operating theatre equipment and supplies are beyond the scope of this publication. Hence, only a few exceptionally critical items are mentioned. Instead, the guidelines state the services that should be provided, with the implication that the training, equipment and supplies needed to provide these services successfully and safely are present. Hence, the following sections 5.4–5.14 of the guidelines list clinical services without division into skills and knowledge versus supplies and equipment.

Head injury is one of the major causes of trauma-related death and disability worldwide. The American Association of Neurological Surgeons (AANS) has developed a set of Guidelines for the management of severe traumatic brain injury (32). These have been shown to improve survival and functional outcome after severe head injury in high-income countries (33, 34). Unfortunately, optimal treatment of head injuries by these protocols requires some of the most expensive resources in the modern therapeutic armamentarium. It is unlikely that low- or even middle-income countries will be able to meet these guidelines fully. In the current guidelines, we have attempted to delineate the most effective diagnostic and therapeutic capabilities that are likely to be achievable at a reasonable and sustainable cost in low- and middle-income countries. It is acknowledged that full compliance with the AANS guidelines would be the most desirable.

Within the resource limitations of developing countries, the assessment of neurological status, including determination of level of consciousness using the Glasgow coma scale, recognition of lateralizing signs, and determination of pupillary size and reflexes, are considered essential at all levels of the health care system, in all countries. This requires only training and perhaps a source of artificial light such as a pocket torch. One of the most significant therapeutic modalities that needs to be applied broadly worldwide is the minimization of secondary brain injury through the maintenance of cerebral perfusion and oxygenation. Most (65%) of the mortality from head injury is associated with secondary brain injury resulting from hypoxia and hypotension (35). This reinforces the primary importance of the ABC outlined in sections 5.1–5.3 above. Recognition of the importance of these factors in patients with head injuries is deemed essential at all hospital levels.

A refinement on the above is to offset the propensity towards raised intracranial pressure (ICP) by avoiding overhydration, principally in haemodynamically stable patients. This knowledge and the understanding that head-injured patients with hypovolemia also require appropriate hydration to prevent hypotension is deemed essential at all hospital levels.

Intracranial pressure (ICP) monitoring for appropriate indications (e.g. Glasgow coma scale less than 9 and abnormal CT scan of the head) and the ability to treat raised ICP through such means as sedation, osmotic diuresis (with mannitol), paralysis, cerebrospinal fluid (CSF) drainage and hyperventilation are deemed desirable at the tertiary care level. They are also desirable at the specialist-level hospital if a neurosurgeon is present. They are also desirable at the specialist-level hospital if a general surgeon with considerable neurosurgical expertise is available, in a setting in which facilities for referral to a tertiary centre are limited.

Intracranial mass lesions with pressure effect account for only around 10–20% of comatose patients. However, timely decompression of these lesions significantly improves outcome. Treatment of these lesions is greatly facilitated by the availability of computerized tomography (CT). This is deemed desirable at all hospital levels. However, its high cost prevents it from being considered essential at any level. CT scans are indeed available in many locations, including low-income countries, but many factors preclude their ready availability to all patients with suspected intracranial mass lesions. These include cost, and in some cases associated mandatory fees, as well as prolonged periods of breakdown (15). Some countries may decide to make CT scanning essential in their own plans. This requires not only the physical presence of the machine, but also timely 24-hour availability to all severely head-injured patients, without regard to ability to pay. It also includes facilities for maintenance and rapid repair within 24 hours. In addition, basic quality improvement programmes should assure that all patients

warranting CT scan of the head (generally Glasgow coma scale of 8 or less) are promptly scanned (generally within 2 hours of arrival to the hospital).

Surgical treatment of intracranial mass lesions is classified as basic (burr hole) or advanced (including craniotomy, craniectomy, treatment of intracerebral haematoma, etc.). CT scans facilitate such treatment, but they are not mandatory. It should be noted that increased survival with drainage of intracranial haematoma was widely documented in the era before CT scans (36). Relief of raised ICP from intracranial mass lesions by burr hole alone implies the skill to perform the operation and the drills or other suitable equipment needed. Some GP-staffed hospitals are situated in isolated places with minimal facilities for timely referral. In these locations, burr holes may reasonably be considered to be “possibly required”. This would imply that a GP with suitable surgical experience would be authorized to perform them. Specialist-level hospitals in most low-income countries would not usually be expected to have a neurosurgeon. In these cases, especially if referral time to tertiary care facilities is prolonged, the ability to perform burr holes by general surgeons is deemed desirable. Moreover, many tertiary care hospitals in low-income countries do not have neurosurgeons. At these facilities, the capability to perform burr holes should be essential. More advanced neurosurgical procedures are deemed “possibly required” at specialist-level hospitals. These would certainly be required if a neurosurgeon were present. However, they would also be reasonably performed by general surgeons in cases where referral to tertiary hospitals was significantly restricted. They would be desirable in any tertiary care environment. However, the dearth of neurosurgical expertise in low-income countries prevents them being deemed essential, even at the tertiary level.

A particular subset of neurosurgical procedures, elevation of open depressed skull fractures, is considered as possibly required at some very isolated GP hospitals, desirable at specialist facilities and essential at tertiary hospitals. Elevation of closed depressed skull fractures is less urgent and hence deemed possibly required at specialist-level hospitals (unless a neurosurgeon is present) and desirable at tertiary hospitals.

The AANS guidelines indicate that steroids are of no proven benefit in the treatment of traumatic head injury. Hence, they are not promoted in the *Guidelines for essential trauma care*.

Finally, malnutrition in head trauma patients has been associated with worsened outcome (32). Maintenance of at least baseline caloric and protein requirements should be assured, including NG feeding if the patient is comatose. This is deemed essential for all head-injured patients with altered neurological status at all hospital levels. This also applies to patients with a prolonged inability to eat, whether for head injury or other forms of trauma.

TABLE 13 **Diagnosis and monitoring**

Resources Monitoring	Facility level			
	Basic	GP	Specialist	Tertiary
Stethoscope	E	E	E	E
Blood pressure cuff	E	E	E	E
Torch (flashlight)	E	E	E	E
Thermometer	E	E	E	E
Foetal stethoscope	D	E	E	E
Urinary catheter with collection bag	D	E	E	E
Electronic cardiac monitoring	I	D	D	D
Pulse oximetry	I	D	D	D
Central venous pressure monitoring	I	D	D	D
Right heart catheterization	I	I	D	D
Intracranial pressure monitoring	I	I	D	D
Radiological investigations				
Plain films	D	D	E	E
Portable plain films	I	D	D	E
Contrast radiography (barium, gastrograffin)	I	I	D	D
Ultrasound for trauma (haemoperitoneum)	I	D	D	D
CT ¹	I	D	D	D
Angiography	I	I	D	D
Image intensification/fluoroscopy	I	I	D	D
MRI ²	I	I	D	D
Nuclear medicine	I	I	D	D
Laboratory tests				
Haemoglobin/haematocrit	D	E	E	E
Glucose	I	E	E	E
Gram stain	I	D	E	E
Bacterial cultures	I	D	D	E
Electrolytes (Na, K, Cl, CO ₂ , BUN ³ , creatinine)	I	D	D	D
Arterial blood gas measurements	I	D	D	D
Serum lactate	I	I	D	D
Other				
Paediatric length-based tape (Broselow tape)	D	D	D	D
Otoscope	D	E	E	E
Ophthalmoscope	D	D	E	E
Compartment pressure measurement	I	D	D	E

¹ CT: Computerized axial tomography.² MRI: Magnetic resonance imaging.³ BUN: blood urea nitrogen.

at GP hospitals that care for a specified minimum number of trauma cases. Given the importance of basic X-ray, the reader is referred to existing guidelines on the WHO Basic Radiology System (BRS) (49). The BRS describes the minimum equipment needed for basic X-ray services, oriented towards small hospitals.

The capability for portable plain X-rays (mobile radiography units) is deemed essential at tertiary care hospitals and desirable at all other hospitals. Other imaging capabilities, including CT scans, angiography, fluoroscopy, image intensification, MRI and nuclear medicine exams, add value to the management of injured patients. They are listed as desirable at various levels, as indicated in the table. Their costs prevent them from being designated essential at any level at this time.

Basic, general-purpose ultrasound is of great utility for non-trauma purposes, especially obstetrics. Hence, one might reasonably wish to assure its availability at all specialist and tertiary hospitals, as well as many GP hospitals, for general purposes. However, ultrasound for trauma requires more advanced skills in both performance and interpretation, primarily for the diagnosis of haemoperitoneum. It is this skill and capability that are considered desirable (rather than essential) at all hospital levels. See also section 5.7 (Management of abdominal injury) for more details.

The measurement of haemoglobin concentration or haematocrit by any suitable, reliable technique is deemed essential at all hospital levels, as is the measurement of serum glucose concentration. Gram stains are essential at specialist and tertiary hospitals. Bacterial cultures are essential at tertiary facilities. Measurements of electrolytes, arterial blood gases (ABG) and serum lactate are deemed desirable. Their costs prevent them from being deemed essential at any level at this time.

The use of a paediatric length-based tape (Broselow tape) is of benefit in the calculation of doses of fluids and medications for children. This is inexpensive and desirable at all institutions that care for injured children. Ophthalmoscopes and otoscopes are useful adjuncts for the physical diagnosis of injured patients and are deemed essential or desirable as indicated in the table. Equipment for the measurement of compartment pressures can be purchased as a ready-made set or can be constructed using tubing and the gauge from a blood pressure cuff. Expertise to use either one is deemed essential at tertiary care facilities.

Any capabilities for monitoring, radiology or laboratory services that are deemed essential or that are converted from desirable to essential in a national plan should meet certain criteria, in addition to mere physical availability of the relevant equipment. These include prompt availability (24 hours a day, 7 days a week, if indicated), sufficient personnel skilled in performing the procedures or tests safely and accurately and in interpreting the results, and, where relevant, sufficient quality assurance mechanisms to monitor the application of the test or procedures. Diagnostic imaging and laboratory facilities should follow WHO

guidelines, including those for internal quality control and external quality assessment (http://www.who.int/eh/ Main_areas_of_work/DIL/About_DIL.htm).

Equipment should be maintained so as to assure the availability of the related services without interruption due to malfunction. This point is to be emphasized, as the ability to manage and maintain medical equipment has often lagged behind acquisition of the equipment (49).

5.14 Safety for health care personnel

Give the high rates of HIV and other bloodborne pathogens worldwide, it is imperative that health workers be provided with adequate protection. This is generally true, but especially so in trauma care, where blood and other body secretions are abundant and where the hectic nature of resuscitation may lead to an increased propensity to contact. Capabilities for universal precautions are deemed essential at all levels of the health care system. This includes the training necessary to apply these. It also implies provision of the necessary materials, including gloves and goggles, as well as capabilities for the safe disposal of sharps and biological materials. All of these are considered essential at all levels of the health care system. In the course of trauma resuscitation, large volumes of fluids may sometimes come into contact with health workers' skin. In such cases, adequate water-resistant clothing/gowns should be available. These are essential at all hospital levels. They are listed as desirable at the basic level, as some such facilities may not care for severely injured patients under such scenarios. Finally, capabilities for anti-HIV, post-exposure prophylaxis with anti-retroviral medications is deemed desirable at all levels.

Throughout these guidelines, numerous invasive procedures have been discussed. Implicit in all of these is that they be conducted under appropriate sterile conditions. The capability to achieve sterility is likewise implicit whenever these procedures are mentioned.

TABLE 14 **Safety for health care personnel**

Safety for health care personnel	Facility level			
	Basic	GP	Specialist	Tertiary
Training in universal precautions	E	E	E	E
Gloves	E	E	E	E
Goggles	E	E	E	E
Sharps disposal	E	E	E	E
Biological waste disposal	E	E	E	E
Gowns	D	E	E	E
Post-exposure prophylaxis for HIV	D	D	D	D

5.15 Special considerations for children

Almost all of the preceding sections require some amendment for children. These amendments concern particular skills in handling injured children, equipment in paediatric sizes, and adjustment of doses of medications. A brief summary of these issues follows, emphasizing items that might need to be amended to further optimize care of the injured child. These apply to items that have already been designated as either essential or desirable.

Airway management (see also section 5.1)

Skills:

- recognition of the differences in airway anatomy in children
- somewhat different techniques needed, particularly for endotracheal intubation

Equipment:

- paediatric sizes for nasal and oral airways, bag–valve–masks, laryngoscopes and endotracheal tubes

Management of respiratory distress (see also section 5.2)

Equipment:

- paediatric-size equipment for oxygen face masks and for chest tubes

Management of shock (see also section 5.3)

Skills:

- knowledge of different baseline vital signs by age
- knowledge of varying physiological responses to blood loss and varying manifestations of shock in children of different ages
- knowledge of paediatric doses for fluids, both for baseline requirements and for treatment of shock
- knowledge of paediatric doses for blood transfusion for treatment of haemorrhagic shock
- skills in the insertion of paediatric intravenous cannulas, in peripheral cutdown access and in insertion of intraosseous lines

Equipment in paediatric sizes:

- intravenous cannulas
- blood pressure cuff
- urinary catheters
- nasogastric tubes
- intraosseous needle or equivalent
- weighing scale

Laboratory facilities:

- ability to perform laboratory tests on small samples of blood from paediatric patients

Head injury (see also section 5.4)

Skills:

- ability to calculate modified Glasgow coma scale for young children

Extremity injury (see also section 5.8)

Skills:

- understanding of specific paediatric orthopaedic injuries which are highly prone to disability
- management of fractures specific to the paediatric age group (e.g. epiphyseal fractures)

Spinal injury (see also section 5.9)

Skills:

- knowledge of the varying anatomy of the childhood spine
- interpretation of spinal X-ray films (required for both non-surgical and surgical management)

Equipment:

- C-collars in paediatric sizes

Burns and wounds (see also section 5.10)

Skills:

- assessment of percentage body surface area of burn wounds in young children

Rehabilitation (see also section 5.11)

Skills:

- monitoring of growth and development to assure that the normal milestones are met as closely as possible, despite the injury and any related physical impairment.

Pain control and medicines (see also section 5.12)

Skills:

- knowledge of paediatric doses

Equipment:

- appropriate references or charts to calculate paediatric doses

Diagnosis and monitoring (see also section 5.13)

Equipment:

- urinary catheters in paediatric sizes
- laboratory capabilities for paediatric volumes (as noted above)
- paediatric length-based tape (Broselow tape) for estimating paediatric doses

6. Methods for promoting essential trauma care services

This section will review some of the methods that could be utilized to promote the implementation of services for essential trauma care. These are methods that have been utilized to promote improvements in medical care generally worldwide. With specific reference to trauma care, they have primarily been used in developed countries. This experience is briefly reviewed. Where information exists on the applications of these methods to trauma care in the setting of developing countries, this is likewise indicated, with thoughts on how such methods might be expanded.

6.1 Training for trauma care

All of the components of these guidelines require skill on the part of the practitioners. Some of these are skills at all levels of the health system and by all levels of providers, such as the skills for initial evaluation and resuscitation. Some are only for hospital workers and some are highly specialized, such as those for operative management. In this manual, we have concentrated on skills for the main care providers, including doctors and nurses. However, it must be recognized that a variety of other professionals are involved in the care of the injured patient. Their skills and training must be considered, including laboratory technicians, X-ray technicians, operating theatre personnel and village health workers. Optimizing the trauma-related skills outlined in these guidelines is a major way in which the EsTC standards can be assured in a cost-effective manner.

Greater attention is needed worldwide to define and optimize the training of doctors and nurses in trauma care. This pertains to both trauma-related skills imparted in basic education and those acquired during postgraduate training, such as house-officer posts and specialist training for doctors. The tables in this manual outline the core set of trauma-related skills that are needed. It is hoped that by so doing this manual will provide guidance for curriculum development in schools of medicine and schools of nursing.

Higher-level skills for operative care require specific training. In rural, low-income environments, most notably in Africa, there is often a need for general practitioners to perform a variety of surgical procedures. In such environments,